

“ RADIOACTIVITY IS... ACCOMPANIED BY CHEMICAL CHANGES IN WHICH NEW TYPES OF MATTER ARE... PRODUCED. ”

Ernest Rutherford, *Philosophical Magazine*, 1902

Radium salt left on photographic plate shows strong radioactivity after development—the yellow tracks are emitting alpha particles.

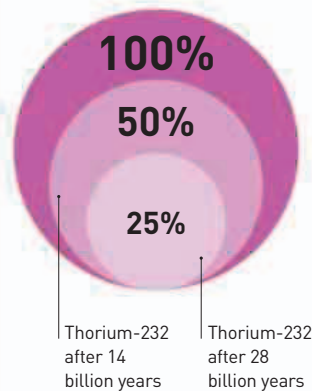
This year also saw the origins of a **revolution in biology**: several biologists rediscovered the **laws of inheritance** that had been established by Gregor Mendel (see 1866). Dutch botanist **Hugo de Vries** (1848–1935) found that the inheritance of characteristics in plants followed rules dictated by particles he called **pangenes** (later changed to genes).

Austrian biologist **Karl Landsteiner** (1868–1943) proposed a theory about **blood group compatibility** in a footnote of a scientific paper. He had found that if the serum (the liquid part of blood) from one person was mixed with another's entire blood, it could cause clumping of the red blood cells. This explained why some blood transfusions were fatal.



De Vries in his garden

Hugo de Vries experimented with breeding plants—as Gregor Mendel had done years before. Although he retired in 1918, he continued his research until his death.



IN FEBRUARY AND AUGUST, BRITISH ENGINEER Hubert Cecil Booth (1871–1955) filed patents for a device that sucked air through a filter system. His invention was the **first powered vacuum cleaner**. In November, American electrical engineer **Miller Reese Hutchinson**, (1876–1944), inspired by a friend rendered deaf by scarlet fever, **patented the first electrical hearing aid**. In a modification of Alexander Bell's telephone technology, Reese Hutchinson's device transmitted sound from a microphone to the ear via a set of headphones.

Ernest Rutherford and British physicist Frederick Soddy (1877–1956) found that **radioactive elements changed into other forms when they emitted radioactivity**. This transmutation, as they later called it (see 1916), always

Radioactive decay

The half-life of a radioactive element is the time taken for a particular element to decay into another form. Thorium-232 has a half-life of 14 billion years.

happened in the same way: for instance, thorium changes into radium. Rutherford identified the time it takes for **half the radioactive material to decay into another form**, which he later **named its half-life**. Soddy went on to demonstrate that some elements had variants, known as isotopes, which may or may not be radioactive. Half-lives of elements and their isotopes vary: for some isotopes of beryllium it is a fraction of a second, but for the element bismuth-209, it is a billion times

longer than the estimated age of the Universe.

In biology, **Karl Landsteiner elaborated on his theory of blood compatibility**. On November 14, he announced that he had identified **three different blood groups, A, B, and O**, on the basis of compatibility patterns. Another rarer blood group, AB, was discovered later.

A meeting of the Zoological Society of London reported on **the discovery of a spectacular new large mammal** from the forests of Africa. **The okapi** had been **discovered by explorer Harry Johnston** (1858–1927) and was described on the basis of examination of its skin and skull.

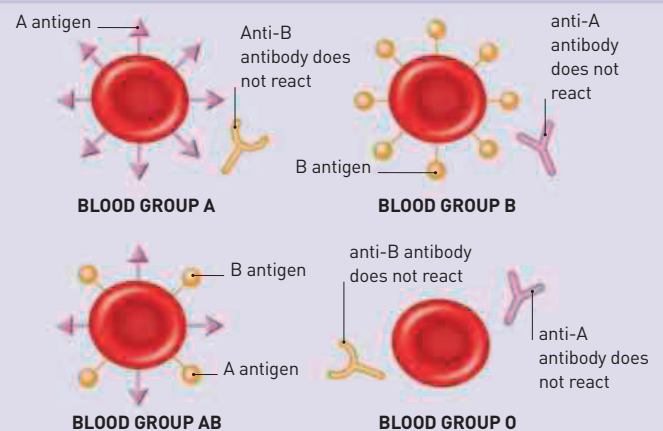
In November, German psychiatrist **Alois Alzheimer**

(1864–1915) examined a woman exhibiting signs of a **severe form of dementia**—and described symptoms of the disease that eventually carried his name. Following her death in 1906, Alzheimer examined her brain and observed the **abnormal plaques** that are characteristic of **Alzheimer's disease**.

On December 12, Italian inventor **Guglielmo Marconi** was reported to have **sent the first radio signal across the Atlantic Ocean**—from Porthcurno, the most southwesterly tip of England, to Newfoundland, in North America. Although some people suggested that it was nothing more than interference, others described it as a deliberate Morse code signal.

BLOOD GROUPS

Red blood cells from some people carry surface components called antigens. Karl Landsteiner identified two types—A and B—which can trigger the fatal clumping (agglutination) if blood is given to someone with appropriately sensitive antibodies. Blood group O has no antigens so it can be donated to anyone. Blood from group AB has both antigens, and can only be given to a person with AB blood.



October Harry Johnston sends skin fragments of a new African mammal to London for identification—it was later named an okapi

January 18 American zoologist Thomas Montgomery describes meiosis (cell division)

May British biologist Ray Lankester confirms okapi is a forest giraffe

November 15 US Patent Office registers a portable electrical hearing aid for Miller Reese Hutchinson

November 26 Alois Alzheimer describes the earliest recorded case of Alzheimer's disease

Late 1901 Ernest Rutherford and Frederick Soddy start to formulate their theory of radioactive decay